# SAURABH KUMAR PANDEY

Senior Applied Scientist, Microsoft

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Summary

Applied Scientist with 4+ years of experience building evaluation frameworks and LLM-judge systems for agentic AI, and finetuning SLMs for edge deployment. Shipped evaluation infrastructure for Word Agent by translating research into production-grade solutions, with publications at top-tier ML and NLP conferences.

Education

- **Indian Institute of Technology, Kharagpur** — Kharagpur, India
  Master & Bachelor of Technology - Dual Degree | GPA: 9.10/10.00 — 2017 - 2022
- **Lucknow Public College** — Lucknow, India
  Intermediate - 98.25% | High School - 93.20% — 2013 - 2016

Work Experience

- **Senior Applied Scientist, Microsoft NLX Applied Science Team** — **Oct 2025 - Present**
  Project: The Word Benchmark (TWB) - Large-scale evaluation of Word Agent
  - Built a dataset of 500+ documents for 20+ agentic tasks, establishing evaluation for Copilot Word Agent
  - Architected a lawyer-judge evaluation framework, boosting pass rate by 8% for long context documents
  - Led judge calibration for 6 LLM judges and 5 agentic workflows; improved inter-judge agreement by 0.11

  Project: Small Language Models (SLMs) for Fluency in Word Editor
  - Fine-tuned an on-device, extreme weight-sharing EdgeFormer model on 1M+ data points for GEC task
  - Engineered an ONNX runtime pipeline for edge devices, achieving a 70 MB memory and 100 ms latency
  - Achieved SOTA for 10M-parameter models with scores of 0.7825 on CONLL-14 and 0.5321 on BEA-40k
  - Evaluated SLMs, including MAI, Phi-4, and Qwen-3.8 as lightweight critique verification judges for GEC

  Project: Memory vs Grounding Arbitration in Agentic Systems
  - Developed a dataset pipeline generating 5,000+ synthetic memories and grounding facts for 11 scenarios
  - Designed a suite of 5 LLM judges to measure document properties, with human-judge agreement of 0.83
  - Evaluated 10,000+ user prompts to guide memory consumption and grounding behavior in Word Agent
- **NLP Research Engineer, MBZUAI, Abu Dhabi, UAE** — **Jul 2024 - Sep 2025**
  Project: Culturally Yours - Cultural Evaluation and Alignment of LLMs
  - Developed an AI cultural reading assistant using LLMs (GPT-4o) to identify CSI spans in review texts
  - Finetuned Llama-3.1 using LoRA and improved alignment score of models by 16% in CSI identification
  - Evaluated knowledge topologies in LLMs using logits based spectral analysis of Llama-3.1 and Gemma-2
- **Machine Learning Engineer, V-Labs, Bengaluru, India** — **Jun 2022 - Jun 2024**
  Project: DExTr - Information Extraction from Documents using LLMs
  - Implemented a RAG-based QnA system for querying complex documents (claims), powered by GPT-3.5
  - Orchestrated an AzureOCR & LLM powered IE module for extracting information from Accord forms
  - Finetuned SLMs using optimized PEFT techniques like LoRA, Q-LoRA, and Prefix Tuning (Acc - 0.62)
  - Improved the accuracy of the extraction pipeline by 15% through suitable document-agnostic prompts

  Project: Check Fraud - Fraud Detection and Prevention using VLMs and LLMs
  - Extracted information on 10 fields from Bank Checks using Transformer architecture with 87% accuracy
  - Integrated ViT encoder and BART decoder as a pipeline architecture to perform Information Extraction
  - Implemented Siamese based Multi-Scale Residual Fusion Network (F1-0.62) for Handwriting Verification
  - Proposed FasterRCNN based Chargrid algorithm to improve localization F-score by 13.45 on documents
  - Trained Donut model for extraction of 5 key-value pairs from invoice documents with an accuracy of 93%

Internships

- **Machine Learning Intern, mavQ (MTX Group), Hyderabad, India** — **May 2021 - Jul 2021**
  - Developed an object detection model (mAP - 0.7292) using an affirmative ensemble of YOLO-v3 & v4
  - Created Flask APIs and trained 30+ model variations implementing YOLO, RetinaNet & EfficientDet
  - Developed ArcFace loss Face Recognition model (Accuracy - 0.95) on LFW, CFP-FP & AgeDB datasets

Technical Skills

- **Languages**: Python, C/C++, SQL
- **Tools**: Microsoft Azure, AWS, Docker, Git, ONNX, Slurm, MCP, PostgreSQL, pgvector, FAISS, Amazon MTurk
- **Frameworks**: PyTorch, TRL, PEFT, LangChain, LangGraph, CrewAI, ScikitLearn, NLTK, TensorFlow, Flask
- **Publications**: EMNLP 2026, ICML 2026, NAACL 2025, COLING 2025, EACL 2024, EMNLP 2023, PNAS 2023